

**FILTERING AND ENHANCEMENT OF MEDICAL SIGNALS USING
TRADITIONAL TECHNIQUES**

*A project report submitted in partial fulfillment of the requirements for the award of the
degree of*

BACHELOR OF TECHNOLOGY

in

Department of Electronics and Communication Engineering

by

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "Filtering and enhancement of medical signals using Traditional techniques. " is a bonafide record of the work done by

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ABSTRACT

In this project, the primary objective is to filter medical signals, particularly Electrocardiogram (ECG) signals, from DAISY Database. The Daisy database typically contains recordings of cardiac activity from both the abdominal and thoracic during pregnancy. Electrodes are placed on the thoracic and abdominal areas of the mother to monitor the cardiac movements of both the mother and fetus. However, a common challenge in such recordings is the interference of maternal electrocardiogram (MECG) signals with the fetal ECG signals. The maternal cardiac activity often overlaps or masks the fetal heartbeat signal, making it challenging to accurately detect and analyze the fetal ECG. Our objective can be done by both traditional as well as non- traditional techniques. To fulfill our objective, we have used traditional techniques such as Butter-worth filtering and Principal Component Analysis (PCA).

KEYWORDS: Electrocardiogram, Principal Component Analysis

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ABBREVIATIONS

ECG	Electrocardiogram
MECG	Maternal Electrocardiogram
PCA	Principal Component Analysis
MSE	Mean Square Error

1 INTRODUCTION

The Electrocardiogram (ECG) stands as a cornerstone in the realm of medical diagnostics, offering invaluable insights into the intricacies of cardiac function. The significance of the ECG lies in its ability to provide clinicians with a real-time depiction of the heart's electrical impulses. By placing electrodes strategically on the body's surface, the ECG captures these electrical signals, offering a comprehensive snapshot of the heart's performance. This non-invasive test has revolutionized cardiovascular medicine, enabling timely detection, diagnosis, and monitoring of cardiac disorders ranging from arrhythmias to myocardial infarctions.

At its core, the ECG manifests as a graphical representation of voltage fluctuations over time, corresponding to the cardiac cycle's various phases. Each distinct waveform and interval within the ECG tracing hold diagnostic significance, offering clues regarding the heart's rhythm, conduction pathways, and overall health. Among these, the heart rate emerges as a pivotal metric, serving as a fundamental indicator of cardiac function and physiological well-being. So it is very essential to filter medical signals which can be done using both traditional and non-traditional techniques. But we have used traditional techniques such as butter worth filtering and Principal Component Analysis.

2. METHODOLOGY

Firstly, we have referred research papers done by various scholars. The data utilized in this study was taken from the Database for the Identification of Systems (Daisy). The database comprises ECG signals from an eight-electrode pregnant lady, with the first five signals being abdominal and the final three being thoracic. We have downloaded this database from google. The last three electrode signals have information where the MECG is dominant. For this first review, our aim is too clean MECG from 6th electrode reading using Butterworth filter.

DATABASE USED:

The data utilized in this study was taken from the Database for the Identification of Systems (Daisy). The database comprises ECG signals from an eight-electrode pregnant lady, with the first five signals being abdominal and the final three being thoracic. The sampling frequency is 250 Hz, and the sample time is four seconds, yielding a signal of 1000 samples, as shown in Fig. 1. The last three electrode signals have information where the MECG is dominant. Nevertheless, the FECG signals are still affecting the MECG, resulting in high-frequency noises in which the QRS information is still readable, but missing information on P and T waves. The high-frequency interference makes the MECG noisy and hence digital filters are required.

Butterworth Filter:

Butterworth filters are characterized by the property that the magnitude characteristic is maximally flat at the origin, and there are no ‘ripples’, either in the pass band or in the stop band.

The transfer function of an nth order Low-pass Butterworth filter is given by:

$H(S)=k_o/(s-s_1) (s-s_2) \dots(s-s_n)$, where K_o is normalizing constant.

Some features of the Butterworth filters represented by equation are:

- They are all-pole filters
- The poles lie on the left half of the s-plane
- They have the 3 dB cut-off at $\omega_C = 1$ rad/sec.

Butterworth filters are one of the most widely used types of analog and digital filters in various applications such as signal processing, communication systems, and control systems. They are known for their simplicity and desirable frequency response characteristics

The key features of Butterworth filters are:

Maximally Flat Magnitude Response: - The magnitude response of a Butterworth filter is maximally flat in the passband, meaning it has no ripple or peaking. This property makes Butterworth filters well-suited for applications where a smooth frequency response is desired, such as audio processing and instrumentation.

Monotonic Frequency Response: - The magnitude response of a Butterworth filter monotonically decreases from the passband to the stopband. This results in a smooth transition between the passband and the stopband, without any overshoots or oscillations.

All-Pole Transfer Function: - The transfer function of a Butterworth filter is an all-pole function, meaning it has only poles and no zeros. This simplifies the design and implementation of Butterworth filters, as the filter design can be carried out using only the pole locations.

Pole Locations: - The poles of a Butterworth filter are evenly spaced around a circle in the left-half of the s-plane. The radius of the circle is equal to the cutoff frequency (ω_c) of the filter, and the number of poles determines the order of the filter.

3 dB Cutoff Frequency: - The 3 dB cutoff frequency of a Butterworth filter is always equal to the design cutoff frequency (ω_c). This property makes it easy to design Butterworth filters with a specific cutoff frequency.

These characteristics make Butterworth filters a popular choice for various applications, as they provide a good balance between performance and complexity. They are often used in audio processing, image processing, control systems, and more, where a smooth frequency response and a well-defined cutoff frequency are required.

Butterworth filtering is a common technique used in ECG signal processing for removing noise and unwanted frequency components while preserving the underlying ECG waveform.

Here are the steps how it works:

1.Filter Design: Butterworth filters are a type of linear filter with a maximally flat frequency response in the passband. They are characterized by their order and cutoff frequency. The order of the filter determines how steeply it attenuates frequencies outside the passband. Higher-order filters have steeper roll offs but can introduce more phase distortion.

2.Bandpass or lowpass filtering: In ECG signal processing, Butterworth filters are often used as either bandpass or lowpass filters. A bandpass filter allows frequencies within a certain range (the passband) to pass through while attenuating frequencies outside this range.

This is useful for removing noise and interference while preserving the ECG signal's frequency content. A lowpass filter, on the other hand, attenuates frequencies above a certain cutoff frequency, which is useful for removing high-frequency noise.

3.Filter application: Once the Butterworth filter parameters (order and cutoff frequencies) are chosen, the filter is applied to the raw ECG signal. This is typically done using digital signal processing techniques. The filtered signal will then contain only the frequency components within the desired passband or below the cutoff frequency.

4.Implementation: Butterworth filters can be implemented using various methods, such as direct form, cascade form, or using filter design functions provided by signal processing libraries like SciPy in Python or MATLAB's Signal Processing Toolbox. These functions allow you to specify the filter type (lowpass, bandpass, etc.), order, and cutoff frequencies, and then apply the filter to the ECG signal.

5. Evaluation: After filtering, it's important to evaluate the effectiveness of the filter in removing noise while preserving the ECG waveform. This can be done visually by comparing the filtered signal to the original raw signal and quantitatively by measuring signal-to-noise ratio (SNR) or other metrics.

Overall, Butterworth filtering is a versatile and widely used technique in ECG signal processing for noise reduction and frequency component selection. However, it's important to choose the filter parameters carefully to avoid distorting the underlying ECG waveform or introducing artifacts.

First, we have plotted time and frequency representations of necessary signals. From the frequency spectrum (amplitude response), we have identified some frequency and corresponding amplitude. By assuming orders from 1 to 3 we have found satisfied cut-off frequency by using magnitude of Butterworth filter formula. Then we have made the above stated $H(S)$ into a low pass filter. Then we have converted analog $H(S)$ into digital $H(Z)$ using bilinear transformation. By inverse Z-transform we got time domain representation of filter. We, found 1st order, second order and third order filters and their time domain representation. Finally convolving obtained filters with original signals from database in MATLAB we got cleaned maternal ECG.

The data utilized in this study was taken from the Database for the Identification of Systems (Daisy). The database comprises ECG signals from an eight-electrode pregnant lady, with the first five signals being abdominal and the final three being thoracic. We have downloaded this database from google. The last three electrode signals have information where the MEGG is dominant. For this second review, our aim is to clean MEGG from 7th and 8th electrode readings using Butterworth filter.

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Finally convolving obtained filters with original signals from database in MATLAB we got cleaned maternal ECG.

Principal Component Analysis:

Principal Component Analysis (PCA) is a widely used statistical technique for reducing the dimensionality of data while preserving most of its variance. It achieves this by transforming the original variables into a new set of uncorrelated variables called principal components (PCs). The first principal component captures the direction of maximum variance in the data, with subsequent components capturing the remaining variance in descending order.

Principal Component Analysis (PCA) using eigenvalues and eigenvectors is a technique used to reduce the dimensionality of a dataset by transforming the original variables into a new set of uncorrelated variables called principal components. The PCA following steps :

1.Covariance Matrix Calculation: The first step is to compute the covariance matrix of the dataset. The covariance matrix measures the relationships between every pair of variables in the dataset.

2.Eigenvalue Decomposition: Next, we perform eigenvalue decomposition on the covariance matrix. Eigenvalue decomposition breaks down the covariance matrix into its constituent parts: eigenvalues and eigenvectors.

3.Selection of Principal Components: The eigenvalues represent the variance of the data along the corresponding eigenvectors. Higher eigenvalues indicate higher variance, and hence, more important principal components. We typically sort the eigenvalues in descending order to identify the most significant principal components.

4.Construction of Principal Components: The eigenvectors corresponding to the top k eigenvalues (where k is the desired number of principal components) are selected to form the transformation matrix. This transformation matrix is used to project the original data onto the new principal component space.

5.Projection: Finally, the original data is projected onto the new space spanned by the selected principal components. This yields a reduced-dimensional representation of the data.

For this we have taken 1st reading from Daisy database which is Abdominal ECG, and we converted the signal into matrix form of equal rows and columns. Then we have applied principal component analysis and reconstructed the matrix. We have compared this with 6th, 7th

,8th electrode readings from database after converting them into matrix form. We have taken Mean Square Error of corresponding element in these electrode readings and reconstructed matrix. If the MSE is very less, then we can say that abdominal reading is almost similar to that particular electrode reading that means we have extracted MEG from AEG. The extracted MEG can be further cleaned by filtering techniques. Similarly, we have found regenerated matrix for all the remaining Abdominal ECG signals and found MSE values comparing with MEG readings.

PERFORMANCE MEASURES USED:

The performance measure used in this project is Mean Square Error (MSE).

In the context of Principal Component Analysis (PCA), the Mean Squared Error (MSE) can be used as a performance measure to evaluate the effectiveness of the PCA in reconstructing the required data from its reduced-dimensional representation.

PCA aims to reduce the dimensionality of abdominal data by projecting it onto a lower-dimensional subspace while retaining as much of the original variance as possible. After performing PCA and reducing the dimensionality of the data, one common evaluation approach is to reconstruct the original data using the reduced set of principal components and then compare it to the Maternal ECG readings using MSE.

The steps to calculate MSE in PCA typically involve:

1. Perform PCA on the original data to obtain the principal components.
2. Select a subset of principal components or all principal components that capture most of the variance in the data.
3. Reconstruct the original data using these selected principal components neglecting others.
4. Calculate the MSE between the reconstructed data and the maternal ECG data from database.
5. The MSE in PCA serves as a measure of how well the original data can be represented using a reduced number of principal components and used in our application. A lower MSE indicates that the reduced-dimensional representation is better able to capture the variability present in the original data.

Here's a simplified version of the formula to calculate MSE in PCA:

The goal in PCA is often to find the smallest number of principal components that result in a reasonably low MSE, indicating a good balance between dimensionality reduction and preserving important information from the original data. It subtracts each entity of reconstructed matrix with Maternal ECG readings and taking their square divided by total no of entities.

MATLAB CODE:

CA-1:

```
clc;clear all; close all;
%%
data_ecg= importdata('FOETAL_ECG.dat');
j = data_ecg(:,2);
j1=data_ecg(:,3);
j2=data_ecg(:,4);
j3=data_ecg(:,5);
j4=data_ecg(:,6);
j5=data_ecg(:,7);
j6=data_ecg(:,8);
j7=data_ecg(:,9);
k=j(1:1000);
l=j1(1:1000);
m=j2(1:1000);
n=j3(1:1000);
o=j4(1:1000);
p=j5(1:175);
q=j6(1:175);
r=j7(1:175);
figure
subplot(3,1,1);
plot(p);
title('Electrode6');
subplot(3,1,2);
plot(q);
title('Electrode7');
subplot(3,1,3);
plot(r);
title('Electrode8');
figure
% ylabel('Magnitude');
figure
subplot(3,1,1);
plot(abs(fft(p)));
title('FFT of MEEG Signal from electrode6');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
subplot(3,1,2);
plot( abs(fft(q)));
title('FFT of MEEG Signal from electrode7');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
subplot(3,1,3);
plot( abs(fft(r)));
title('FFT of MEEG Signal from electrode8');
xlabel('Frequency (Hz)');
```

```

ylabel('Magnitude');
figure
figure
num=[63.24 63.24];
den=[65.24 61.24];
n=max(length(num),length(den));
[g t]=impz(num,den,n+1);
%stem(g);
subplot(4,1,1);
plot(q);
title('original 7th electrode ')
z1=conv(g,q);
subplot(4,1,2);
plot(z1);
title('after filtering first order(MECG)')
num1=[494.06 1482.18 1482.18 494.06];
den1=[169.06 203.94 27.14 -7.7];
n1=max(length(num1),length(den1));
[g1 t]=impz(num1,den1,n1+1);
%stem(g1);
z2=conv(g1,q);
subplot(4,1,3);
plot(z2);
title('MECG after filtering second order')
num2=[245.912 737.736 737.73 245.912];
den2=[234.89 -142.71 -108.17 15.99];
n3=max(length(num2),length(den2));
[g2 t]=impz(num2,den2,n3+1);
%stem(g);
z3=conv(g2,q);
subplot(4,1,4);
plot(z3);title('MECG after filtering third order')

```

CA-2:

```

% Define the one-dimensional signal
%signal = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10];
%data_ecg= importdata('reading1.csv');
clc;
close all;
clear all;
data_ecg= importdata('FOETAL_ECG.dat');
j = data_ecg(:,2);
j1=data_ecg(:,3);
j2=data_ecg(:,4);
j3=data_ecg(:,5);
j4=data_ecg(:,6);
j5=data_ecg(:,7);
j6=data_ecg(:,8);

```



```

j7=data_ecg(:,9);
signal = j4;%aecg reading1
% Define the dimensions of the matrix
num_rows = 50;
num_cols = 50;

% Calculate the total number of elements in the signal
num_elements = numel(signal);

% Calculate the number of elements per column in the matrix
elements_per_column = ceil(num_elements / num_cols);

% Initialize the matrix
matrix = zeros(num_rows, num_cols);

% Fill the matrix with elements from the signal
for i = 1:num_elements
    % Calculate the row and column index for the current element
    row_index = mod(i - 1, num_rows) + 1;
    col_index = ceil(i / num_rows);

    % Assign the element from the signal to the matrix
    matrix(row_index, col_index) = signal(i);
end

%%

% Display the original signal and the resulting matrix
disp('Original signal:');
disp(signal);
disp('Matrix:');
disp(matrix);
%%Find the eigenvalues
[eigenvalues, eign_vet] = eig(matrix);

disp('Eigenvalues of matrix :');
%k=disp(eigenvalues);
%l=disp(eign_vet);
%%Find the magnitude
magnitude = abs(eigenvalues);

disp(['Magnitude of complex numbers = ']);
disp(magnitude);

[x,ind]=sort(diag(magnitude),"descend");
%[x1,ind]=sort(magnitude,"descend");
%eigen_vects=eign_vet(ind,ind);
%eigens=eigenvalues(:,ind);
sortedeigvect=eign_vet(:,ind);

```

```

xx =x/max(x);
k=xx(1:4);%contain principal eigen values
num_components=4;
principal_components=sorteigvect(:,1:num_components);
%D1=k(:,1);V1=principal_components(:,50);
%D1*V1;
lamda=diag(k);
Rematrix=principal_components*lamda;
%%%%%%by sir
[coeff,score,latent,tsquared,explained,mu] = pca(matrix);
K=coeff*score'
%x2 = length(find(xx>=0.7));
% Vs=k(:,ind);
% x1=find(xx>=0.7);
% % y=disp(x1);

% eigen_vects=eign_vet(ind,ind);
% eigens=x1(:,ind);

%%%%6th electrode
signal1=j5(1:2500);
plot(signal1);
% Define the dimensions of the matrix
num_rows1 = 50;
num_cols1 = 50;

% Calculate the total number of elements in the signal
num_elements1 = numel(signal1);

% Calculate the number of elements per column in the matrix
elements_per_column1 = ceil(num_elements1 / num_cols1);

% Initialize the matrix
matrix1 = zeros(num_rows1, num_cols1);

% Fill the matrix with elements from the signal
for i = 1:num_elements1
    % Calculate the row and column index for the current element
    row_index1 = mod(i - 1, num_rows1) + 1;
    col_index1 = ceil(i / num_rows1);

    % Assign the element from the signal to the matrix
    matrix1(row_index1, col_index1) = signal1(i);
end
disp(matrix1);
% Calculate MSE
mse = mean((K(:) - matrix1(:)).^2);

%%%%7th electrode

```

```

signal2=j6(1:2500);
plot(signal2);
% Define the dimensions of the matrix
num_rows2 = 50;
num_cols2 = 50;

% Calculate the total number of elements in the signal
num_elements2 = numel(signal2);

% Calculate the number of elements per column in the matrix
elements_per_column2 = ceil(num_elements2 / num_cols2);

% Initialize the matrix
matrix2 = zeros(num_rows2, num_cols2);

% Fill the matrix with elements from the signal
for i = 1:num_elements2
    % Calculate the row and column index for the current element
    row_index2 = mod(i - 1, num_rows2) + 1;
    col_index2 = ceil(i / num_rows2);

    % Assign the element from the signal to the matrix
    matrix2(row_index2, col_index2) = signal2(i);
end
disp(matrix2);
% Calculate MSE
mse1 = mean((K(:) - matrix2(:)).^2);
%%8th electrode
signal3=j7(1:2500);
plot(signal3);
% Define the dimensions of the matrix
num_rows3 = 50;
num_cols3 = 50;

% Calculate the total number of elements in the signal
num_elements3 = numel(signal3);

% Calculate the number of elements per column in the matrix
elements_per_column3 = ceil(num_elements3 / num_cols3);

% Initialize the matrix
matrix3 = zeros(num_rows3, num_cols3);

% Fill the matrix with elements from the signal
for i = 1:num_elements3
    % Calculate the row and column index for the current element
    row_index3 = mod(i - 1, num_rows3) + 1;
    col_index3 = ceil(i / num_rows3);

    % Assign the element from the signal to the matrix

```

```

    matrix3(row_index3, col_index3) = signal3(i);
end
disp(matrix3);
% Calculate MSE
mse2 = mean((K(:) - matrix3(:)).^2);

% Convert the matrix into a one-dimensional signal
finalsignal1 = matrix(:);
finalsignal2 = matrix1(:);
finalsignal3 = matrix2(:);
finalsignal4 = matrix3(:);
finalsignal5 = K(:);
subplot(5,1,1);
plot(finalsignal1);
title('Original fifth electrode ');
subplot(5,1,2);
plot(finalsignal5);
title('Original fifth electrode reconstructed');
subplot(5,1,3);
plot(finalsignal2);
title('Original 6th electrode');
subplot(5,1,4);
plot(finalsignal3);
title('Original 7th electrode ');
subplot(5,1,5);
plot(finalsignal4);
title('Original 8th electrode')

```

3 RESULTS AND DISCUSSIONS

CA-1:

The graphs obtained after filtering using Butterworth filter for 6th electrode readings are:

ORIGINAL ELECTRODE READING:

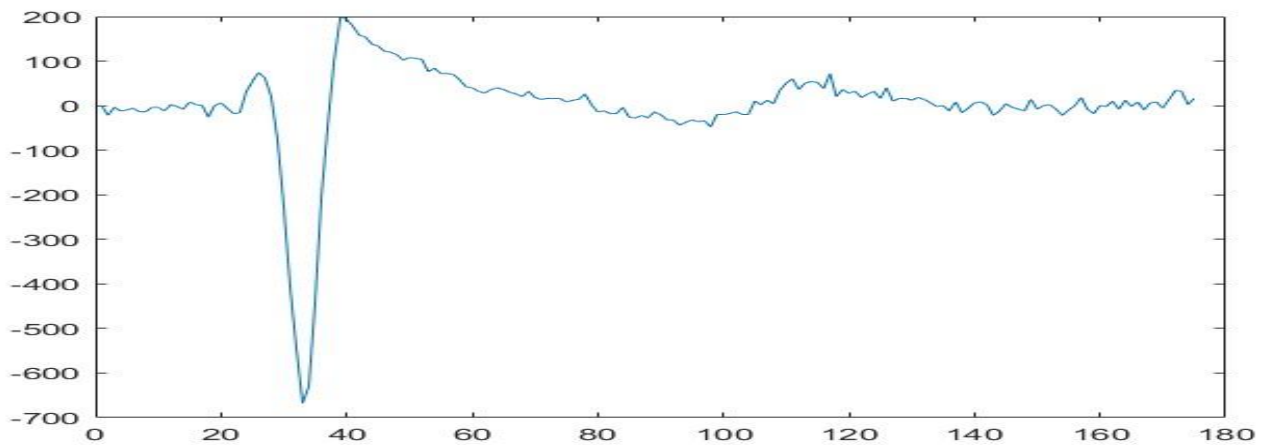


Fig3.1

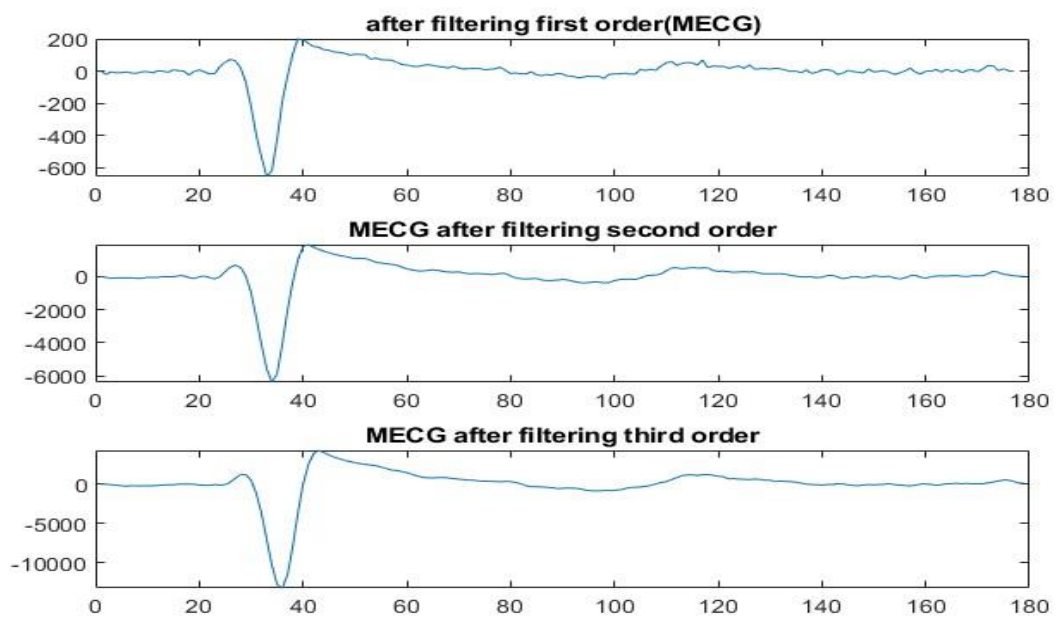


Fig3.2

CA-2:

The graphs obtained after filtering using Butterworth filter for 7th and 8th electrode readings are:

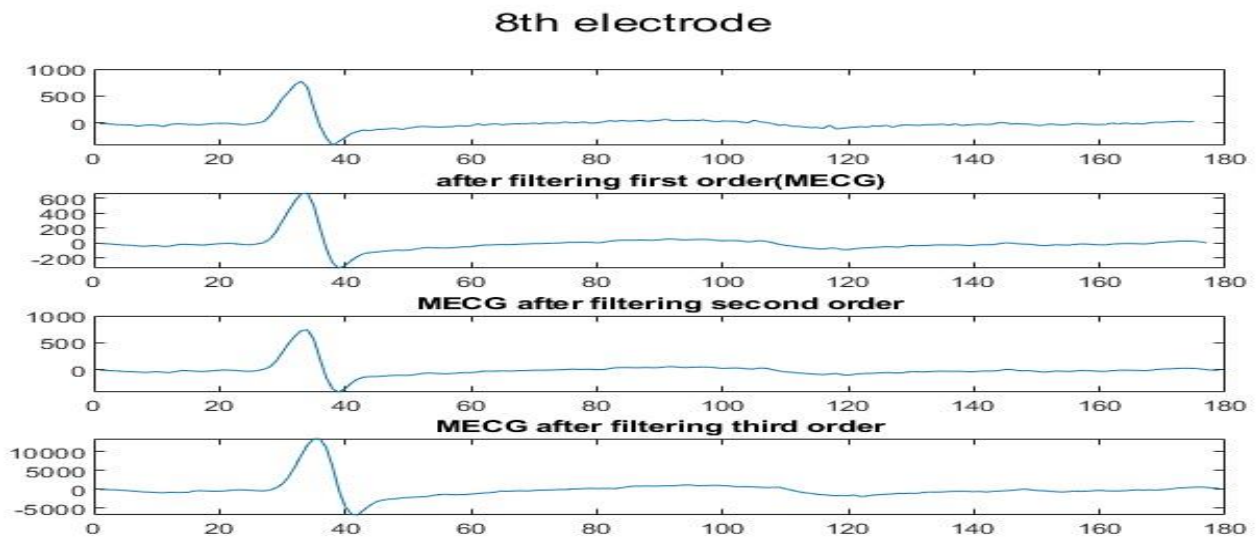


Fig3.3

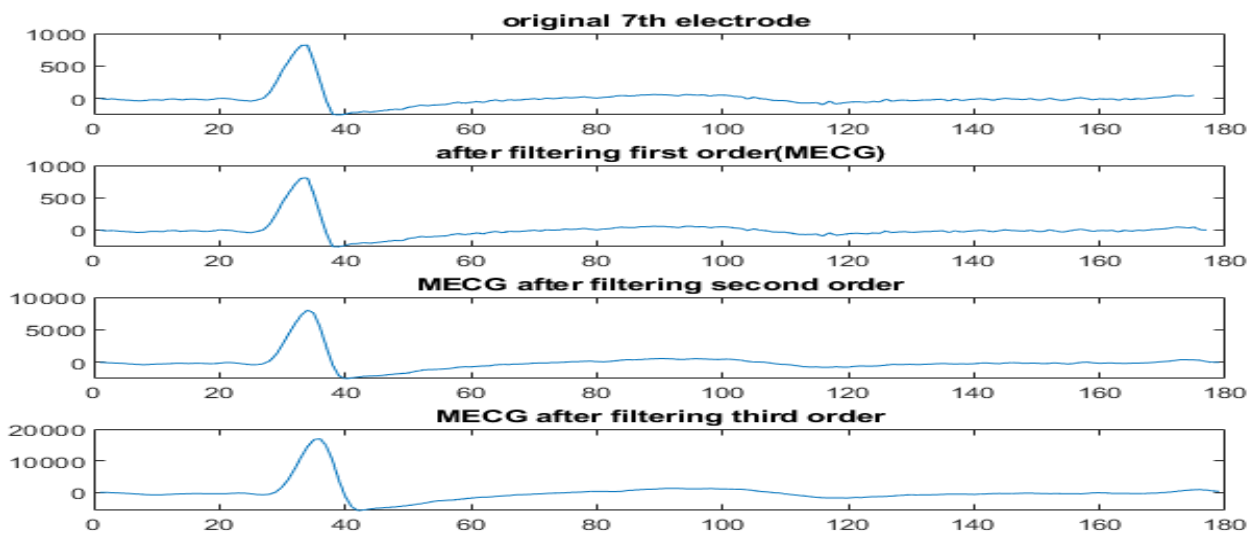


Fig3.4

AFTER CA-2:

PLOTS OF RECONSTRUCTED READINGS AND ORIGINAL R

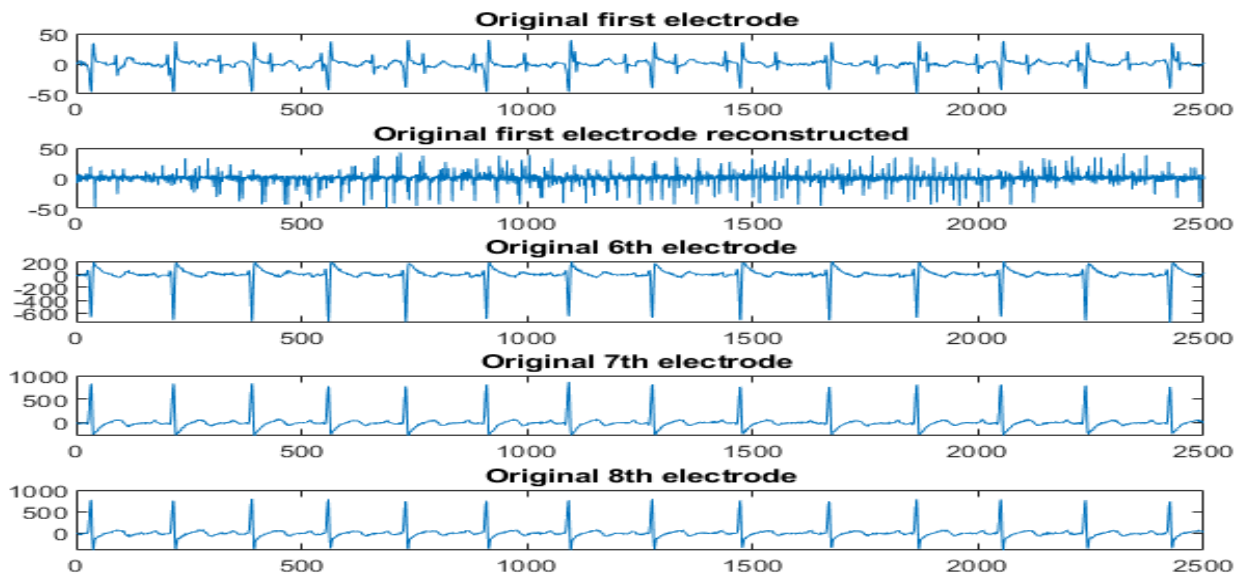


Fig3.5

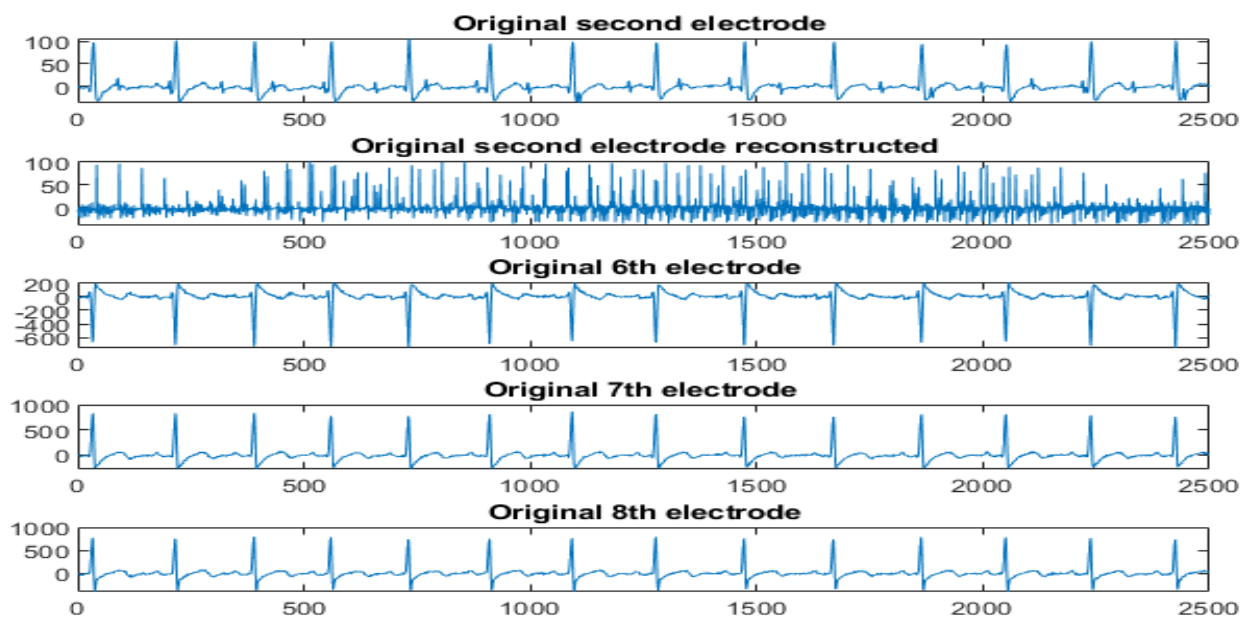


Fig3.6

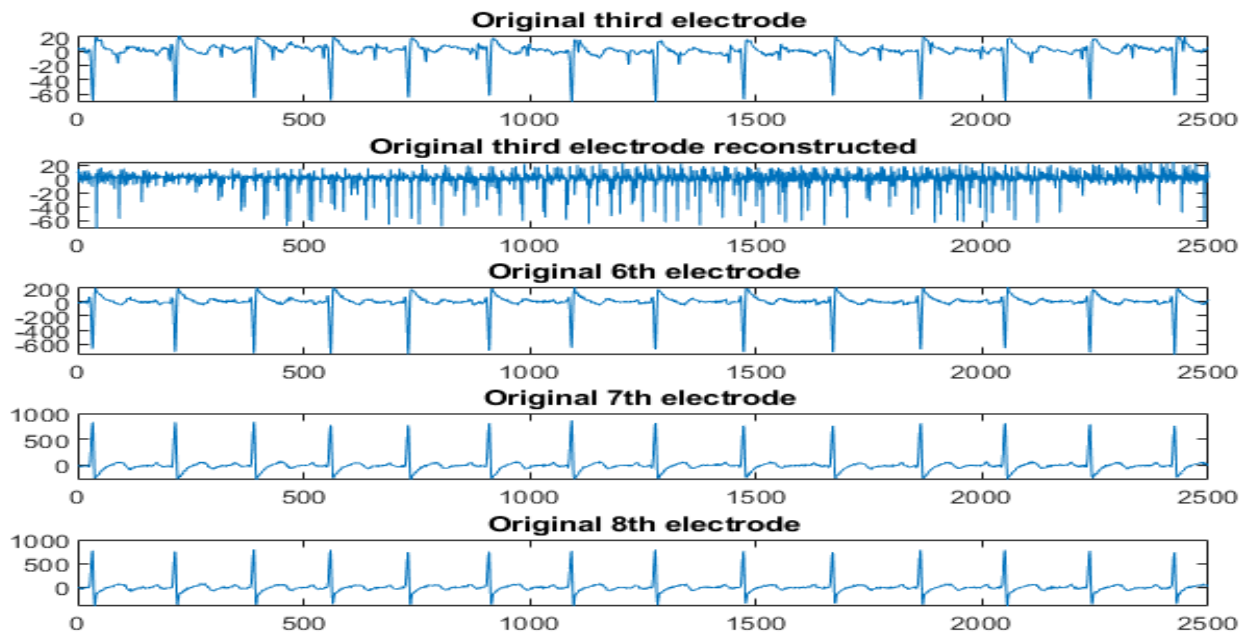


Fig3.7

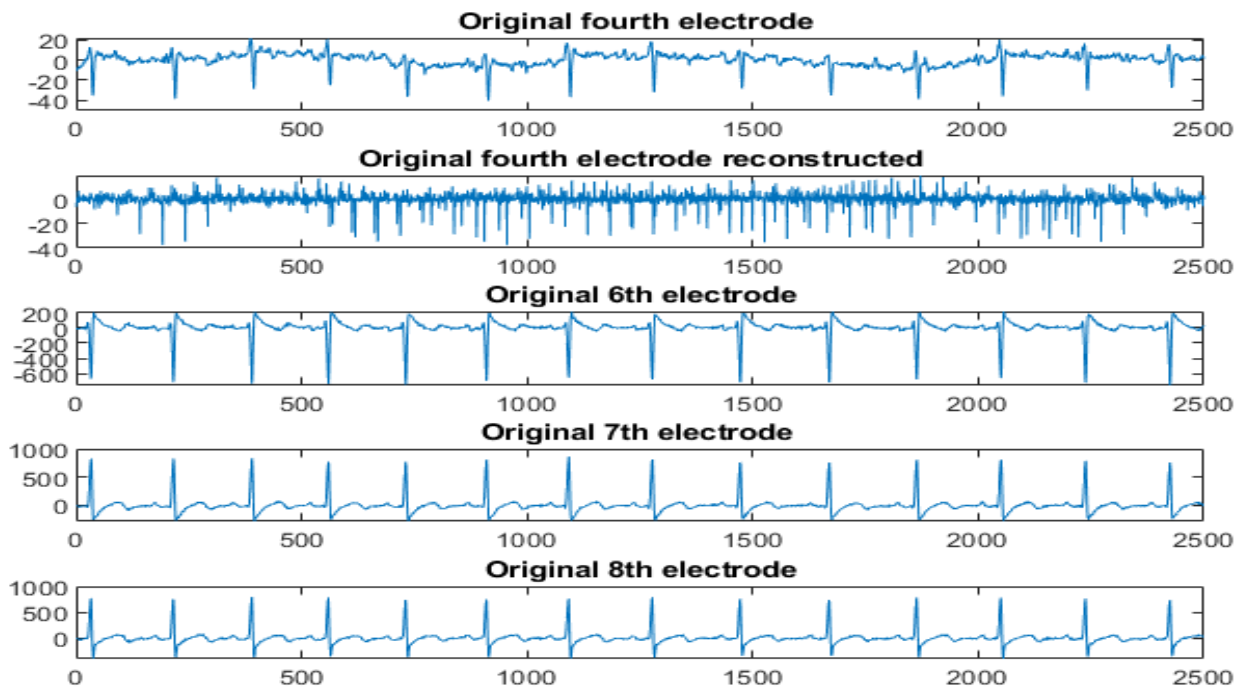


Fig3.8

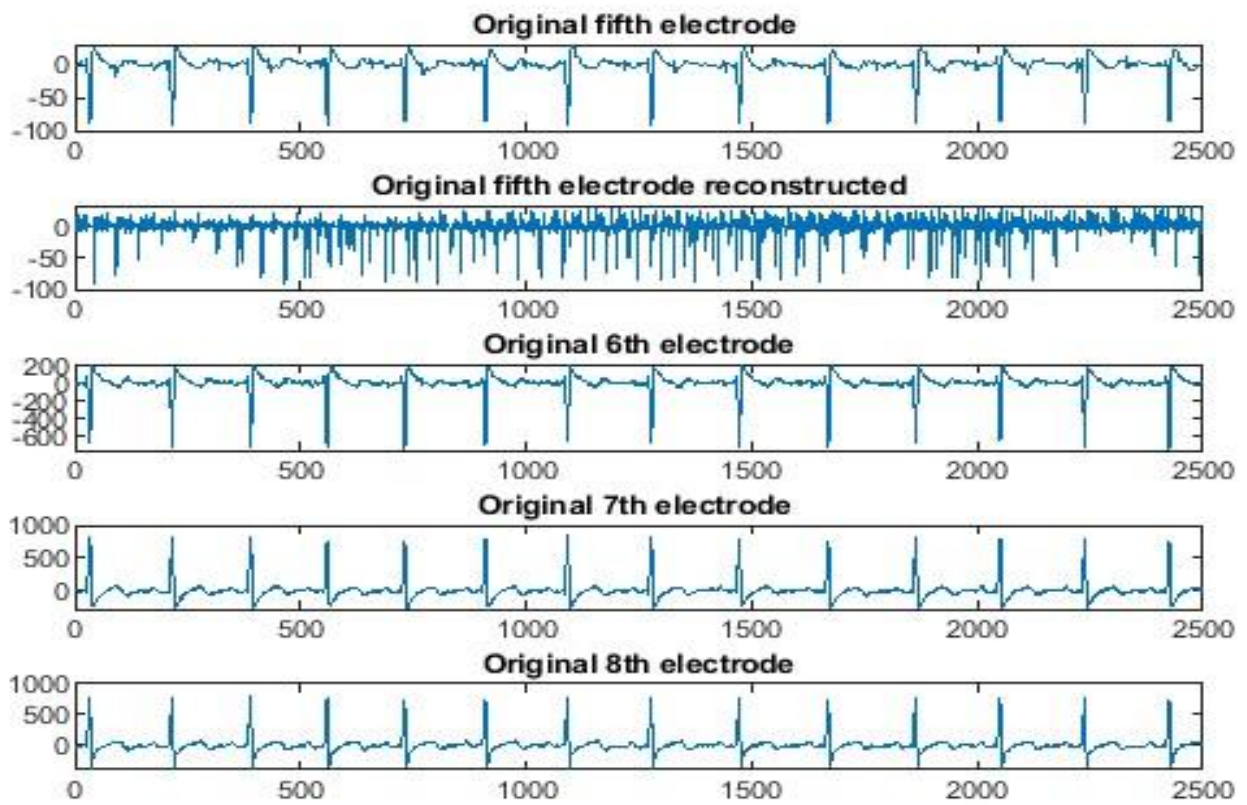


Fig3.9

MSE VALUES OBTAINED AFTER COMPARING REGENERATED ABDOMINAL ELECTRODE READINGS AND MATERNAL READINGS:

ABDOMINAL READINGS	6TH ELECTRODE	7TH ELECTRODE	8TH ELECTRODE
1ST ELECTRODE	1.2407	1.8965	1.2444
2ND ELECTRODE	1.2509	1.9348	1.7072
3RD ELECTRODE	1.2444	1.9032	1.6750
4TH ELECTRODE	1.2317	1.8974	1.6683
5TH ELECTRODE	1.2561	1.9109	1.6823

From the above table 3.1 it can be observed that MSE obtained by comparing regenerated matrix from abdominal electrode readings and 6th electrode so it can be concluded that by suitable techniques we can get Fetal Electrocardiogram by subtracting AECG from 6th electrode.

COMPARISION BETWEEN BUTTERWORTH FILTERING AND PCA

Filtering of ECG signals using Butterworth filtering and Principal Component Analysis (PCA) serves different purposes and approaches:

Butterworth Filtering:

Purpose: Butterworth filtering aims to remove noise and unwanted frequency components from the ECG signal while preserving the underlying waveform.

Approach: Butterworth filters are designed based on specific filter characteristics such as order and cutoff frequency. They operate in the frequency domain and attenuate frequencies outside the desired passband. Butterworth filters are effective for removing noise but do not inherently extract features or reduce dimensionality.

Advantages: Effective in removing noise and interference. Simple to implement using digital signal processing techniques.

Disadvantages: May not preserve all relevant information in the signal, especially if the filter parameters are not chosen carefully. Does not explicitly extract meaningful features from the data.

Principal Component Analysis (PCA):

Purpose: PCA is a dimensionality reduction technique that aims to capture the underlying structure of high-dimensional data by projecting it onto a lower-dimensional subspace spanned by orthogonal vectors called principal components.

Approach: PCA identifies the directions of maximum variance in the data and projects the data onto these directions, thereby reducing its dimensionality. In the context of ECG signals, PCA can be used to extract meaningful features or reduce the dimensionality of the data for visualization or further analysis.

Advantages: Can reveal underlying patterns or structures in the data. Provides a compact representation of the data while preserving most of its variance. Useful for feature extraction and data visualization.

Disadvantages: Assumes linear relationships between variables and may not capture nonlinear relationships in the data. The interpretation of principal components may not always be straightforward, especially in complex datasets.

Comparison:

Butterworth filtering and PCA serve different purposes in ECG signal processing. Butterworth filtering primarily focuses on noise reduction, while PCA focuses on dimensionality reduction and feature extraction. Butterworth filtering operates in the frequency domain and removes unwanted frequency components, while PCA operates in the feature space and identifies the most informative directions in the data. Butterworth filtering is typically applied as a preprocessing step to remove noise before further analysis, while PCA can be used for feature extraction, dimensionality reduction, or data visualization.

In some cases, both techniques can be used together in a pipeline, with Butterworth filtering applied first to remove noise, followed by PCA for dimensionality reduction or feature extraction.

APPLICATIONS:

Clinical Applications:

Maternal ECG can be used to:

1. **Monitor Cardiac Function:** Assess the mother's cardiac health and detect any abnormalities, such as arrhythmias or ischemic changes.
2. **Detect Fetal Distress:** Changes in the maternal ECG may be indicative of fetal distress, which can prompt further investigation and intervention.
3. **Guide Obstetric Management:** Maternal ECG can inform decision-making regarding the

timing and mode of delivery, particularly in cases of known or suspected cardiac complications.

Principal Component Analysis (PCA) is a widely used technique in the analysis of electrocardiogram (ECG) signals. ECG signals contain a wealth of information about the heart's electrical activity, and PCA can be a valuable tool for extracting and analyzing the most important features of these signals.

Maternal ECG is typically performed using standard 12-lead electrodes placed on the mother's chest and abdomen. The resulting ECG waveform is then analyzed for various parameters, such as:

1. **Heart Rate:** Maternal heart rate typically ranges from 60-100 beats per minute during pregnancy.
2. **PR Interval:** The time between the onset of the P wave and the beginning of the QRS complex, which reflects the time taken for the electrical impulse to travel from the atria to the ventricles.
3. **QRS Complex:** The duration and morphology of the QRS complex, which represents the ventricular depolarization.
4. **QT Interval:** The time from the start of the QRS complex to the end of the T wave, which reflects the total duration of ventricular depolarization and repolarization.

In the context of ECG signal analysis, PCA can be applied in the following ways:

1. **Dimensionality Reduction:** ECG signals are typically high-dimensional, with multiple leads and a large number of data points over time. PCA can be used to reduce the dimensionality of the ECG data by identifying the principal components that capture the most significant variations in the signal. This can help in processing and analyzing the data more efficiently, as well as in visualizing the underlying patterns and features.
2. **Feature Extraction:** The principal components obtained from PCA can be used as features for various ECG-based applications, such as arrhythmia detection, heart rate variability analysis, and myocardial infarction diagnosis. These principal components represent the most important characteristics of the ECG signal, which can be more informative and discriminative than the raw ECG data.
3. **Noise Reduction:** ECG signals are often contaminated by various types of noise, such as baseline wander, muscle artifacts, and power-line interference. PCA can be used to identify and remove these noise components, effectively enhancing the signal-to-noise ratio of the ECG data. This can improve the accuracy and reliability of subsequent analyses and applications.
4. **Signal Compression:** The dimensionality reduction property of PCA can also be used for ECG signal compression, which is particularly important for telemedicine and remote monitoring applications where data storage and transmission bandwidth are limited.

5. Visualization: PCA can be used to create low-dimensional visualizations of ECG data, such as scatter plots or 3D plots, which can help in the identification of patterns, clustering, and anomalies in the data. These visualizations can provide valuable insights into the underlying physiological processes and help in the interpretation of ECG signals.

In the specific context of ECG signal analysis, PCA can be applied to various preprocessing steps, such as baseline wander removal, signal denoising, and feature extraction. The principal components obtained from PCA can then be used as inputs to machine learning algorithms for tasks like arrhythmia classification, heart disease diagnosis, and patient monitoring. Overall, the application of PCA in ECG signal analysis has been shown to improve the accuracy, robustness, and interpretability of various ECG-based applications, making it a valuable tool in the field of biomedical engineering and clinical cardiology.

4 CONCLUSION

Electrocardiogram (ECG) signals play a crucial role in the medical field, providing valuable information about the heart's electrical activity. However, these signals are often corrupted by various noises and interferences during the recording process. This is particularly true in the case of maternal ECG (MECG) signals, which are essential for monitoring the health of both the mother and the fetus during pregnancy. In this study, we have focused on the effective cleaning of MECG signals using traditional signal processing techniques, specifically the Butterworth filter. Then we have utilized the Database for the Identification of Systems (DaISy) to obtain the necessary ECG signals from an eight-electrode pregnant woman, where the last three electrode signals contained dominant MECG information.

The successful extraction of the cleaned MECG signals highlights the effectiveness of the Butterworth filter in addressing the challenges associated with noisy ECG recordings. This approach can be valuable in various medical applications, such as monitoring the maternal and fetal health during pregnancy, as well as in the broader context of cardiac condition assessment. Principal Component Analysis (PCA) can be a useful technique for analyzing ECG signals, particularly when dealing with large datasets or when trying to extract meaningful features from the signals. It is also very good traditional approach for ECG signal processing. So finally, it can be concluded that traditional techniques play an important role in filtering of medical signals.

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